THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** living and non-living components interacting with each other	Biotic and abiotic parts together.
2	**(d)** food chain	Linear sequence.
3	**(a)** decomposition	Recycling.
4	**(b)** sunlight	Non-living.
5	**(b)** grass, grasshopper, frog, snake	Producer first.
6	**(b)** carbon dioxide	CO_2 + water.
7	**(c)** bacteria	Living.
8	**(c)** producers	Green plants.
9	**(c)** oxygen	Oxygen.
10	**(c)** producers	Autotrophs.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Community of living organisms interacting with their physical environment (1); Pond, forest (1).	1+1
12	Break down dead matter and recycle nutrients (1); Bacteria and fungi (1).	1+1
13	Variety of living organisms in an area (1); Keeps the ecosystem stable (1).	1+1
14	Biotic: plants, animals (1); Abiotic: air, water, soil, sunlight (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Biotic: algae, fish, frogs, bacteria (1); Abiotic: water, light, temperature, minerals (1); They interact with each other (1).	1+1+1
16	Bacteria convert atmospheric nitrogen to usable form (1); Rhizobium in root nodules (1); Plants use it to make proteins (1).	1+1+1
17	Break down dead plants and animals (1); Return nutrients to soil (1); Keep surroundings clean (1).	1+1+1
	OR: Producers make food (1); Consumers eat producers or other consumers (1); Decomposers recycle nutrients (1).	1+1+1
18	Loss of habitat (1); Soil erosion (1); Reduction in rainfall and rise in CO_2 (1).	1+1+1
	OR: Many trees and large biodiversity (1); Complex food webs (1); Affects rainfall and climate (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Biotic: producers, consumers, decomposers (1); Abiotic: light, water, air, soil (1); Producers need abiotic factors (1); Consumers depend on producers (1); Decomposers recycle matter (1).	1+1+1+1+1
	OR: Food web is disturbed (1); Prey populations rise (1); Vultures: carcasses are not cleaned (1); Disease spreads (1); Balance is lost (1).	1+1+1+1+1
20	Chain: grass, deer, tiger (1); Chain: algae, small fish, big fish (1); Web has many links (1); If one organism is lost, alternatives exist (1); Energy passes at each trophic level (1).	1+1+1+1+1
	OR: Bees pollinate (1); Birds disperse seeds (1); Rhizobium and pulses (1); Lichens: algae and fungus (1); Both benefit (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** many interlinked food chains	Network.
2	**(b)** fixing nitrogen	Nitrogen fixation.
3	**(a)** make food by photosynthesis	They use sunlight, water and CO2.
4	**(c)** deer	Eats producers.
5	**(b)** carbon dioxide	Greenhouse gas.
6	**(b)** herbivore	Plant-eater.
7	**(b)** tiger	Top of the chain.
8	**(d)** transpiration	Loss through leaves.
9	**(b)** omnivore	Eats both.
10	**(a)** from producers to consumers	One way.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Grass, grasshopper (1); Frog, snake (1).	1+1
12	Deforestation (1); Pollution or hunting (1).	1+1
13	Organisms that make their own food (1); All other organisms depend on them for food (1).	1+1
14	A feeding position in a food chain (1); e.g. producer, primary consumer (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Energy enters through sunlight and producers (1); Passes to consumers (1); Lost as heat; cannot be reused by producers (1).	1+1+1
16	Plant trees (1); Reduce, reuse, recycle (1); Avoid pollution (1).	1+1+1
17	Bees pollinate flowers (1); Birds disperse seeds (1); Bacteria in the gut aid digestion (1).	1+1+1
	OR: Organisms use energy in life activities (1); Much energy is lost as heat (1); Only a small part passes to the next level (1).	1+1+1
18	Herbivore: deer (1); Carnivore: lion (1); Omnivore: human, bear (1).	1+1+1
	OR: Waste reduces dissolved oxygen (1); Fishes die (1); Food chain is disturbed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sun is the source of energy (1); Grass (producer) traps it by photosynthesis (1); Energy passes to grasshopper, frog and snake (1); Part is lost as heat at every step (1); So too little energy is left for further links (1).	1+1+1+1+1
	OR: Stable ecosystems (1); Food and medicine (1); Protected areas (1); Laws against hunting (1); Public awareness (1).	1+1+1+1+1
20	Photosynthesis removes CO2 (1); Respiration adds CO2 (1); Decomposition releases CO2 (1); Burning fuels adds CO2 (1); Oceans and forests are sinks (1).	1+1+1+1+1
	OR: Evaporation (1); Transpiration (1); Condensation (1); Precipitation (1); Run-off and groundwater (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **How Nature Works in Harmony** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** loss of habitats and soil erosion	Harmful.
2	**(a)** decomposers	Bacteria and fungi.
3	**(d)** much less	Most of the energy is lost as heat in life processes.
4	**(b)** dumping plastic waste in rivers	Pollution.
5	**(b)** fungi	Fungi.
6	**(a)** much energy is lost at each step	Energy loss.
7	**(c)** creating wildlife sanctuaries	Protected areas.
8	**(d)** Sun	Sunlight.
9	**(d)** parasite	Lives at the cost of the host.
10	**(d)** the natural home of an organism	Where it lives.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Stable condition where populations in an ecosystem remain steady (1); Maintained by food relationships and cycling (1).	1+1
12	Food chain is a single sequence (1); Food web is a network of interlinked chains (1).	1+1
13	Used in life processes and lost as heat (1); Only a small part passes on (1).	1+1
14	Their prey will increase greatly (1); This upsets the ecosystem (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Evaporation from water bodies and transpiration (1); Condensation forms clouds (1); Rain returns water to land and sea (1).	1+1+1
16	Grass to grasshopper and mouse (1); Grasshopper to frog; mouse to snake and hawk (1); Frog to snake; snake to hawk (1).	1+1+1
17	Protect wildlife (1); Preserve habitats (1); Conserve biodiversity (1).	1+1+1
	OR: Plants take CO2 in photosynthesis (1); Animals and plants release CO2 in respiration (1); Balance is maintained (1).	1+1+1
18	Insects increase (1); Plants may be overeaten (1); Snakes may starve (1).	1+1+1
	OR: Aquatic: pond, sea (1); Terrestrial: forest, desert (1); Different abiotic factors (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Nitrogen fixation by bacteria (1); Lightning also fixes nitrogen (1); Plants absorb nitrates (1); Animals get it by eating plants (1); Decomposers return nitrogen to the soil (1).	1+1+1+1+1
	OR: Pond: algae and water plants (1); Forest: trees and grasses (1); Consumers differ (1); Decomposers in both (1); Abiotic factors differ (1).	1+1+1+1+1
20	Deforestation (1); Pollution (1); Overfishing (1); Urbanisation (1); Climate change (1).	1+1+1+1+1
	OR: Reduce (1); Reuse (1); Recycle (1); Plant trees, save water (1); Use renewable energy (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.